



777 - CARRIER PIN EXTRACTING TOOL

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

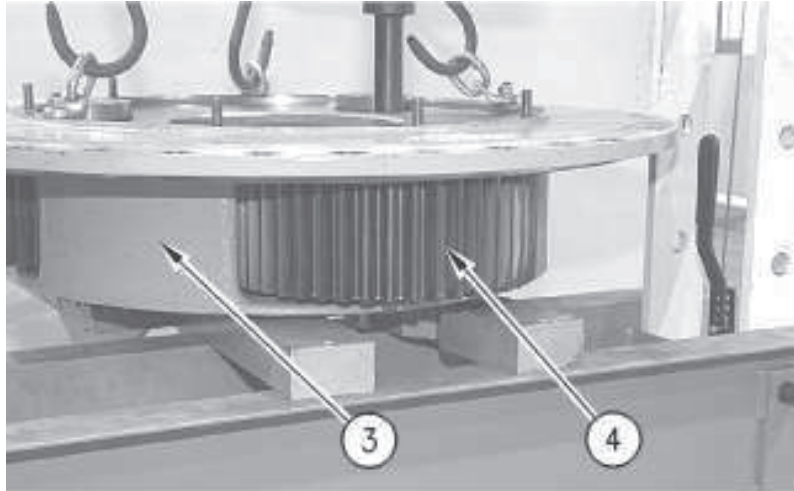
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1.0 Introduction

The correct lifting of the carrier for further disassembly is very important because it allows us to do the job more efficiently and more safely, thus preserving the proper functioning of the service shop.

The current procedure shows the use of three lifting links and chain that form a pyramid to lift the planetary carrier onto a hydraulic press to remove the planetary shaft from the planetary carrier. These chains interfere with the hydraulic press making it impossible to position the planetary carrier correctly to extract the planetary shaft.



This photo was taken from the procedure RENR3366 Final Drive – Disassemble in SIS web. Item 3 shows the planetary carrier and Item 4 is the gear.

2.0 Best Practice Description

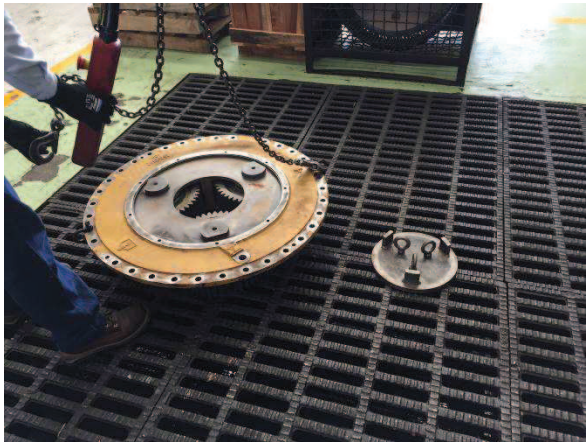
- It would be best to hoist the carrier from the center so that the chain does not interfere with the hydraulic press, thus allowing for adjustments and alignment with the pin for extraction.
- This improvement would make it faster and easier to lift the piece in addition to providing security and allowing workers to complete their work with less of a likelihood of encountering problems or incidents.

3.0 Implementation Steps

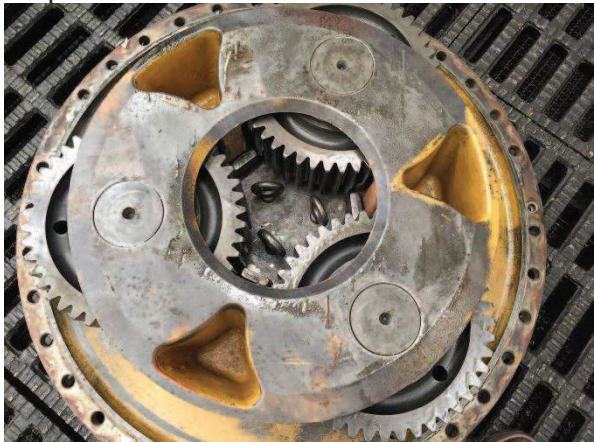
See step by step depiction of the implementation from the images below.

Step 1:

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Step 2:

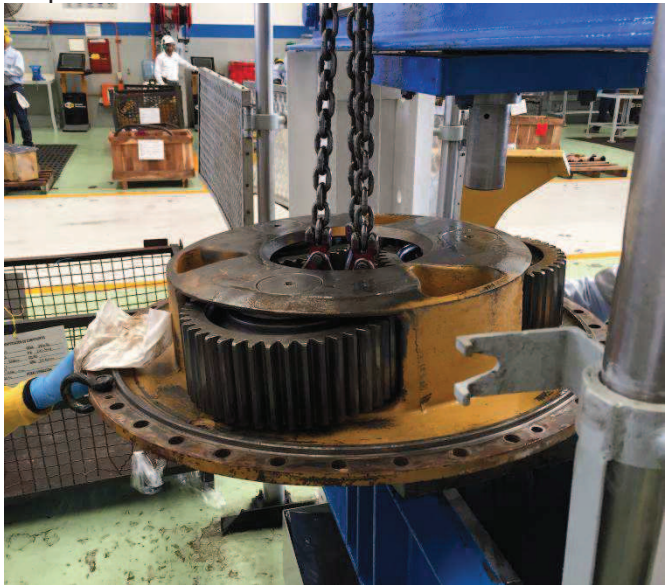


Step 3:

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Step 4:



Step 5:

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Step 6:



4.0 Benefits

This tool provides the following benefits:

- Quick clamping of the part to be readily used as the tool for the job
- Proper alignment of the planetary pins with the stem of the hydraulic press allows for better handling of carrier
- Decreases risk of incidents due to increased stability of the tool

5.0 Resources Required

- 5/8 inch ASTM A36 steel plate, cut in a circle of 320 mm diameter, with three threaded holes in the center for holding the workpiece and three 50mm x 50mm rectangular stops made from the same 5/8 inch ASTM A36 plate welded at 270 mm diameter and aligned with the three threaded holes.
- The tool has a diameter of 320mm for a height of 63.5mm and a weight of 9.44Kg.

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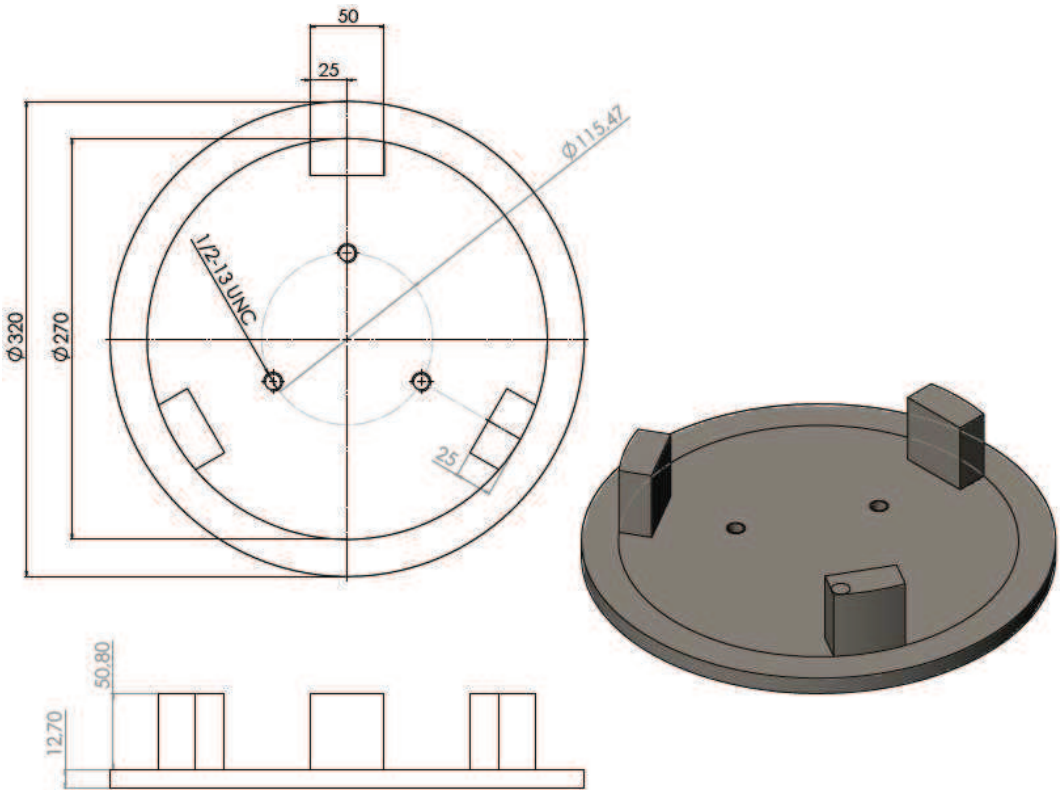
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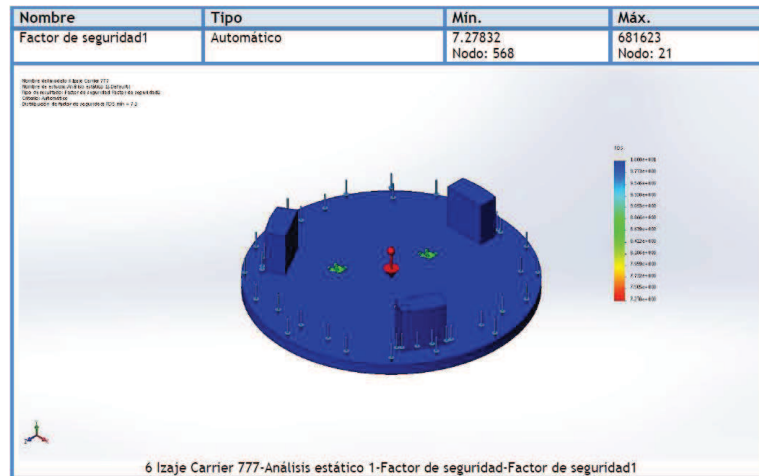


- Materials and equipment required by manufacturing level
- 1 welder

6.0 Supporting Attachments / References



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The minimum safety factor of the design is 7.3, with the maximum stress concentration located in the anchor point on the support plate. With a minimum required safety factor of 5.0, the tool is therefore safe to use. (The load applied to the tool is 150kg / 1480N in the contact face of the tool with the carrier.)

7.0 Related Best Practices

None

8.0 Acknowledgements

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